

Python started as a scripting language to glue together real code, but it's increasingly being used as the main language by many developers. When small devices have enough memory and computational power, developers are free to choose the language that makes their lives easier, and that is often turning out to be Python.

Python is the language of choice for one of the most popular microcontrollers in the market, the Raspberry Pi. There are also versions of boards designed to be even smaller. The Micro Python board and software package is a small microcontroller optimised to run Python on a small board that's only a few square centimetres in area.

The benefits of using Python for IoT: Listed below are some of the advantages of using Python for IoT.

1. It is a very simple language to learn, and is easy to implement and deploy; so you don't need to spend a lot of time learning formatting standards and compiling options.
2. Since it is portable, expandable and embeddable, it is not system-dependent. Hence, it supports a lot of the single-board computers in the market these days, irrespective of the architecture and operating system.
3. Most importantly, it has a huge community which provides a lot of support and libraries for the language.

JavaScript

While many still think of JavaScript as a language for popping up alert boxes on Web pages, its relatively recent popularity on the server makes it a surprisingly favoured choice for IoT applications.

JavaScript is used as the programming language in all the Web browsers and HTML. This is a programming language that shares its libraries with other languages. JavaScript makes things easier as it makes the devices interoperable. Most of the work is focused on the servers and hubs that collect information and then store it.

The smaller smart hubs and sensors that run Linux can usually run Node.js. But even if most of the Node.js code runs on larger machines, there are some ongoing efforts to bring Node.js to smaller machines. Espruino and Tessel are two examples of microcontrollers that run JavaScript from the beginning. Tessel, for instance, is built around Node.js, making it easy for Web developers to move into the IoT without learning a new language.

The benefits of using JavaScript for IoT: Here are some benefits of using JavaScript for IoT.

1. JavaScript has the support of Node.js.
2. It is a robust scripting language which communicates with browsers.
3. There is a vast availability of modern development resources, such as JavaScript libraries, testing tools, prototyping tools, client-side development frameworks, server-side JavaScript APIs, and more.
4. Another important factor that makes JavaScript a

suitable technology for the Internet of Things is the vast availability of learning resources.

PHP

At a time when industries are getting automated, synchronised cloud services and the centralised management of applications can work well with the distributed architecture. The advanced concepts and powerful technology features of PHP can do wonders for the IoT industry. While the language's role on the server, that of juggling micro services, is an obvious application, it is also finding some traction at the lowest level. A number of Raspberry Pi developers are talking about starting up a full LAMP stack with Apache, MySQL and PHP running on top of Linux. They are effectively inverting the paradigm and turning the lowliest thing on the Internet into a full Web server.

If a Raspberry Pi has enough spare cycles—and it often does—then putting a LAMP stack on the chip makes development simpler. All of the server-side code developed over the last 20 years can find a home in a small sensor. PHP may seem like a toy next to the full-strength, pointer juggling machismo of C, but it gets the job done quickly.

The benefits of using PHP for IoT: Listed below are the main advantages of using PHP for the IoT.

1. Easy and simple to learn.
2. Extremely flexible.
3. Easy integration and compatibility.
4. Efficient performance.
5. Cost-efficient. **END** 

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